Thin layer physics

Group N°29: Juan Prieto Alejandro , Kull Jaeggli Jeremy , Yago Viada de Soto

October 17, 2025 Yong Liu

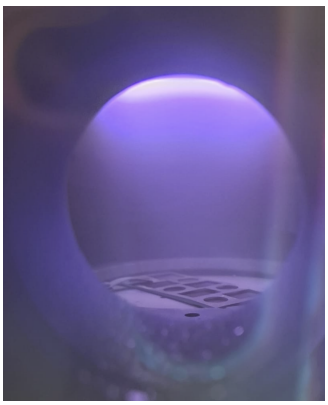


Figure 1: Plasma inside magentron sputtering chamber

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1 Introduction

During the 1990s, the rapid advancement of microprocessor technology and flat-panel displays led to an unprecedented demand for semiconductor materials and expertise. Semiconductors have since become the backbone of modern electronics, enabling applications ranging from computing and communication to renewable energy and sensors [1].

Among the wide family of semiconductor technologies, thin film deposition plays a central role in the development of microelectronic and optoelectronic devices. The controlled growth of nanometer-scale layers allows precise tuning of electrical, optical, and structural properties, which are crucial in devices such as solar cells, transparent conductive coatings, and touch screens [2].

Zinc oxide (ZnO) is a II–VI semiconductor with a direct wide bandgap of about 3.37 eV and a high exciton binding energy of 60 meV at room temperature. These properties make ZnO particularly interesting for transparent electronics, ultraviolet photodetectors, gas sensors, and thin-film transistors. However, the material still presents challenges related to reproducibility, doping control, and defect management [3].

The objective of this laboratory work is to study the physics behind the thin-layer deposition of zinc oxide on glass substrates using magnetron sputtering. Several samples are produced under different experimental conditions, varying the *exposure time*, *in-situ* heating, and post-deposition annealing temperature.

The resulting layers are then characterized according to four main aspects: surface roughness, using profilometry, ,electrical resistivity, optical transmission and reflection, for optical property analysis, and crystal structure, determined by X-ray diffraction (XRD).

This investigation aims to establish the relationship between deposition parameters and the physical properties of ZnO thin films, offering insight into their optimization for optoelectronic applications.

2 Theory

2.1 Cathodic pulverisation magnetron sputtering

Magnetron sputtering is a thin film (μm thick) deposition technique based on the ejection of atoms from a solid target (cathode) under the impact of energetic ions produced in a plasma. A voltage difference V is applied between the target (cathode) and the substrate (anode) in a low-pressure inert gas atmosphere, typically argon. When the applied electric field is strong enough, argon atoms are ionised:

$$\mathrm{Ar} + e^- \rightarrow \mathrm{Ar}^+ + 2e^-. \tag{1}$$

The generated Ar^+ ions are accelerated toward the negatively charged target, causing the ejection of surface atoms via momentum transfer — a process known as sputtering. These atoms then travel through the plasma and condense on the substrate, forming a thin film. The process is easealy visualised in Figure 2

Permanent magnets behind the target generate a magnetic field $\mathbf{B}$ perpendicular to the electric field $\mathbf{E}$. This configuration traps electrons near the target surface through the Lorentz force , increasing the ionisation probability of argon and thus enhancing the plasma density. This leads to higher deposition rates and better film uniformity at lower pressures (typically 10^{-3}–10^{-2} mbar).

The kinetic energy E_k of sputtered atoms is typically a few eV, allowing good adhesion to the substrate without significant thermal damage. The sputtering yield Y, defined as the number of ejected atoms per incident ion, depends on the ion energy and the target material.

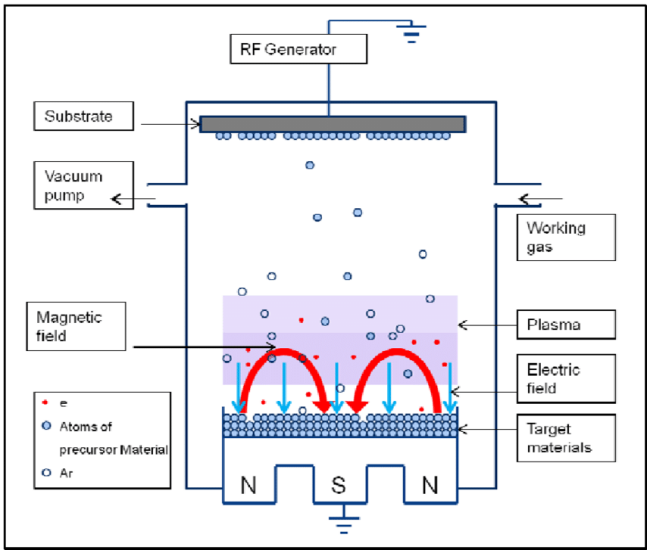


Figure 2: Magnetron sputtering seutp scheme [4]

2.2 Profileometry

Profileometry is a contact measurement technique used to determine the surface topography and film thickness. A stylus with a well-defined radius is scanned across the sample surface, and the vertical displacement $z(x)$ is recorded as a function of lateral position. The film thickness h can be extracted as the step height between the deposited region and the substrate:

$$h(x) = z_{\text{film}}(x) - z_{\text{sub}}. \tag{2}$$

The measurement resolution depends on the profileometry machine parameters such as stylus shape, applied force, and scanning speed. From the obtained profile, one can also compute the root mean square roughness:

$$R_q = \sqrt{\frac{1}{L}\int_0^L [h(x)]^2\, dx}, \tag{3}$$

which quantifies the surface uniformity and thus the quality of the deposition. The considered thickness of the sample is found using the median thickness of the smaple.

2.3 Van der Pauw method for resistivity measurements

The Van der Pauw measuring method, is an experimental setup that allows the determination of the resistivity, ρ, of arbitrarily shaped thin films of uniform thickness t, provided the sample is homogeneous, isotropic, and the contacts are placed on the periphery. Four resistances are measured using successive pairs of contacts, like shown in Figure 3

Following from Ohm's law, the resistance between poles is obtained :

$$R_1 = \frac{V_{34}}{I_{12}}, \qquad R_2 = \frac{V_{41}}{I_{23}}, \qquad R_3 = \frac{V_{12}}{I_{34}}, \qquad R_4 = \frac{V_{23}}{I_{41}} \tag{4}$$

The resistivity is then obtained :

$$\rho = \frac{\pi}{\ln 2}\bar{R}dF\left(\frac{R_1}{R_2}\right), \tag{5}$$

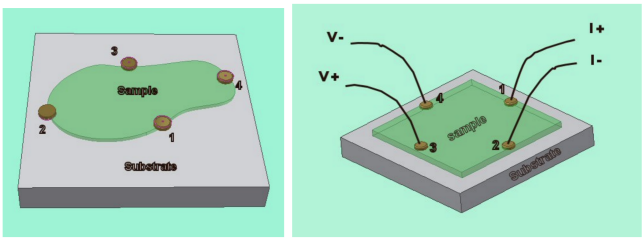


Figure 3: Scheme of Van der Pauw resistivity setup [5]

where $\bar{R}$ is the average of the four resistance measurements in eq. (4).

$$\bar{R} = \frac{1}{4}\sum_{i=1}^{4} R_i, \tag{6}$$

d is the Zinc film thickness and $F\left(\frac{R_1}{R_2}\right)$ is a correction factor with values between 0 and 1, which takes into account the symmetry of the sample. In general, and for this specific stufy shapes like circles or squares one can consider $F = 1$.

2.4 X-ray diffraction

X-ray diffraction (XRD) is used to characterise the crystalline structure of thin films. When an incident X-ray beam of wavelength λ interacts with atomic planes separated by a distance d, constructive interference occurs for specific angles θ satisfying Bragg's law:

$$2d\sin\theta = n\lambda, \tag{7}$$

where n is the diffraction order. Measuring the intensity of the reflected beam as a function of 2θ allows identifying the crystal structure, lattice constants, and preferred orientations of the deposited material.

Peak broadening can also provide information about crystallite size D, according to the Scherrer equation:

$$D = \frac{K\lambda}{\beta\cos\theta}, \tag{8}$$

where β is the full width at half maximum (FWHM) of the diffraction peak and K is a shape factor (typically $K \approx 0.9$).

2.5 Optical properties

The optical characterisation of thin films provides information about their transparency, refractive index, and energy band gap. When light of intensity I_0 passes through a film of thickness d, part of it is absorbed or reflected, leading to a transmitted intensity I_d given by Beer–Lambert's law:

$$I_t = I_0 e^{-\alpha d}, \tag{9}$$

where α is the absorption coefficient.

In the high absorption region, the optical transmission of the compound sample and glass T_{comp} can be expressed :

$$T_{\text{comp}}(\lambda) = \frac{A}{B}e^{-\alpha(\lambda)d} \tag{10}$$

where $A = 16n_1^2 n_2$,$B = (1 + n_1)^3(n_1 + n_2)^3$, n_1 and n_2 the refraction indices of the sample and substrate.

Then from eq. (10) one can derive the wave-length dependence of the absorbtion coefficient :

$$\alpha(\lambda) = \frac{1}{d} \ln \left(\frac{A}{B} \frac{1}{T_{comp(\lambda)}} \right) \tag{11}$$

This coefficient can be related to the film's band structure. For semiconductors such as ZnO, near the absorption edge the dependence of α on the photon energy $h\nu$ follows the Tauc relation:

$$(\alpha h\nu)^n = C(h\nu - E_g), \tag{12}$$

where C is a constant, E_g is the optical band gap, and n depends on the nature of the electronic transition ($n = 1/2$ for direct allowed transitions, $n = 2$ for indirect ones).

By plotting $(\alpha h\nu)^n$ versus $h\nu$ (Tauc plot), the band gap E_g can be extracted as the x-intercept of the linear region of the plot.

In addition, the thickness of the sample can also be estimated from the interference fringes in the transmission spectrum. In the transparency region, the wavelengths λ_1, λ_2 at two consecutive extrema are used to compute the thickness :

$$d = \frac{1}{2n_1} \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1} \tag{13}$$

3 Experimental protocol

Depositions : The machine used for the Zinc Oxide sputtering is shown in Figure 10 in annex. The specific vacuum chamber is shown in Figure 11

To perform a correct deposition, carry out a two-stage pump-down of the chamber. Once the base pressure reaches and holds at aproximately 6×10^{-5} mbar for about five minutes, introduce argon (Ar) and partially close the turbomolecular-pump valve to maintain a constant, continuously replenished Ar working pressure. Ignite the plasma for 5 min with the shutter (shield) in the closed position in order to clean the target from potential impurities before the deposition. Once everything is heated up and running — i.e., thermal and plasma conditions are stable — begin the deposition by opening the shutter and sustaining the plasma for the desired duration.

The same protocol is applied when heating samples *in situ*; however, all target temperatures must first be reached and confirmed by the thermocouple before igniting the plasma.

Once the deposition is done, the pump is shut off and allowed to slow down,and finally air can slowly be let in. This process is fully automatized from pump shut off.

3.1 Roughness and thickness measuring

A profilometer is used to determine the thin shit thickness overall. The used parameters for the machines are the tip applied force of $3.0\mu g$ the specific values of the parameters are given in section 7. The profile data is then analysed statistically to determine thickness and roughness, like shwon in Eq. 3

3.2 Optical measurements

The protocol for measuring Transmittance and Reflectance of a sample is straightforward. First, the baseline is set by measuring the transmittance of air, or the reflectance of a Silicon sample

(depending on what we are measuring), then background values are established by closing an internal shutter, so that no there is output from the light beam, and the background noise is measured. Finally, the sample is placed under the light source, the cylindrical shutter is lowered around it (see Fig.9 in annex), and the measurement is taken.

The light source is a Halogen lamp which radiates light between 350nm to 3500nm. However our measurements only go up to 1100nm (to cover the visible spectrum which is mostly the region of interest).

3.3 X-ray diffraction

In order to analyze the Zinc film samples by X-ray diffraction, it was necessary to cut one of the squares or circles visible in Fig. 4 off the sample. It was then fixed with a type of clay to a metal container and it was made sure that the sample was horizontal. The program set for the machine, measured the intensity of the refracted light for an angle 2θ from 10° to 80°.

3.4 Sample generation

A total of 7 samples were made for this experiment, each with different set parameters. All made samples and correspondent parameters are shown in Tab. 1

3.5 Annealing process

The annealing process is done in heat ovens. The oven is preheated to the desired temperature. Afterwards, the sample is placed on a crucible and inserted in the oven with the adequate heat protection.

4 Results

The ZnO films fabricated for this study were deposited under different parameters: sputtering times, substrate temperatures, and annealing conditions. Profilometry provided the thickness and surface roughness, while the electrical properties were determined using the Van der Pauw method. For a clearer analysis, samples are labeled in as shown in Tab 1 .

Parameter	**S0.1**	**S0.2**	**S1.1**	**S1.2**	**S2.1**	**S3.1**	**S4.1**	**S5.1 / S5.2**
Exposure Time	30 min	1 h	30 min	1 h	30 min	30 min	30 min	30 min / 1 h
Deposition Temp. (°C)	20	20	20	20	20	200	400	20
Annealing Temp. (°C)	–	–	200	200	400	–	–	–
Annealing Time	–	–	60 min	60 min	60 min	–	–	–

Table 1: Summary of fabricated samples with deposition and annealing parameters.

Each deposition has separated squares and circles made by the cache. To simplify organization, the labeling code followed is shown in figure 4 below. "Sq" stands for a measurement performed on a square and "C" stands for "circle". The number indicates which column is used.

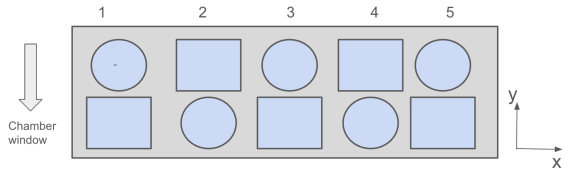


Figure 4: Sample labeling

4.1 Profile measurements

Using the profilometer, the profiles of all samples shown in Fig. 13 in annex were obtained. From this measures, the median thickness can be easily determined, and the surface roughness can be calculated using Eqs. (2) and (3), as presented in Table 2. For sample $S4$, two measurements were performed since the deposition was not uniform across the sample, as can be observed in Fig. 12 in annex.

Sample	**S1.1**	**S1.1 Annealed**	**S1.2**	**S1.2 Annealed**
Thickness d [μm]	0.83 ± 0.15	0.74 ± 0.13	1.57 ± 0.24	1.61 ± 0.11
Roughness R_q [μm]	0.147 ± 0.006	0.128 ± 0.005	0.246 ± 0.010	0.121 ± 0.004

Sample	**S2.1**	**S2.1 Annealed**	**S3.1**	**S4.1.C1**	**S4.1.C5**
Thickness d [μm]	0.48 ± 0.05	0.35 ± 0.04	0.94 ± 0.07	0.49 ± 0.02	0.82 ± 0.04
Roughness R_q [μm]	0.053 ± 0.002	0.036 ± 0.001	0.072 ± 0.002	0.016 ± 0.001	0.045 ± 0.001

Table 2: Thickness and roughness of samples

4.2 Sample resistivity /conductance

Following Van der Paw resistivity measruement method, the conductance of each sample is calculated and is shown in Table3, alongside the realtive errors of used variables in equation 5

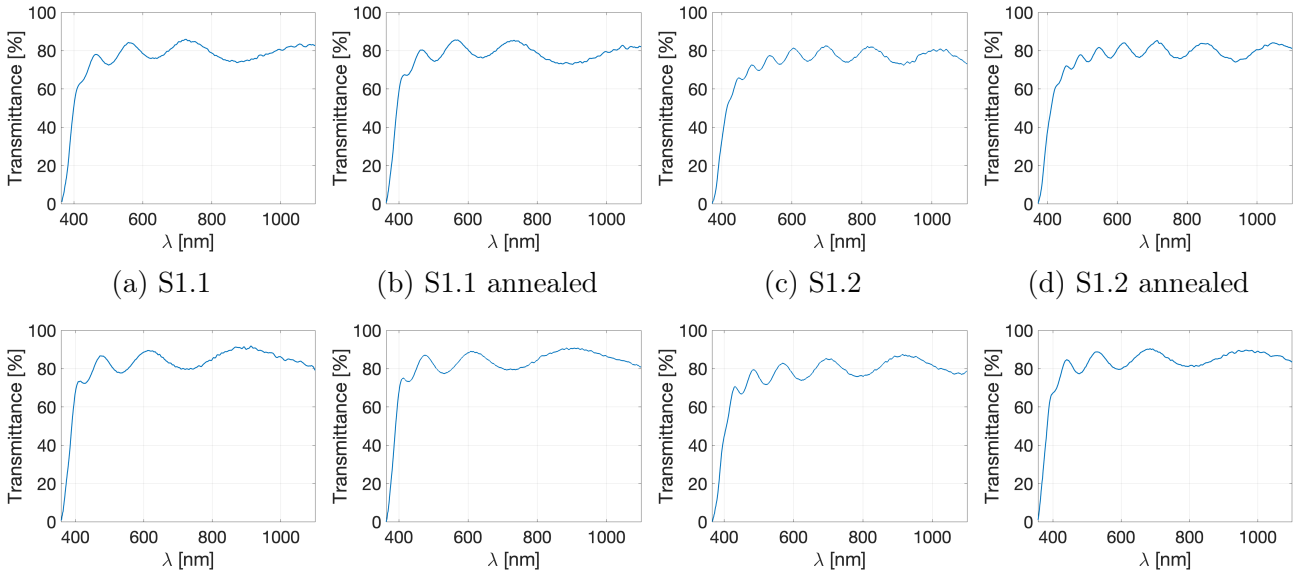
Sample	$R \pm \Delta R$ [Ω]	$\rho \pm \Delta\rho$ [Ω m]	$\Delta\rho/\rho$
S1.1 (30 min)	(2743.23 ± 0.32)	$(4.89 \pm 0.89) \times 10^3$	0.18
S1.1 Annealed	(175.53 ± 0.02)	$(2.77 \pm 0.05) \times 10^2$	0.18
S1.2 (1 h)	(116.60 ± 0.01)	$(1.17 \pm 0.02) \times 10^2$	0.15
S1.2 Annealed	(91.52 ± 0.01)	$(9.15 \pm 0.06) \times 10^1$	0.07
S2.1 (Before ann.)	(6218.66 ± 0.72)	$(1.56 \pm 0.16) \times 10^3$	0.10
S2.1 Annealed	(277.08 ± 0.03)	$(2.77 \pm 0.03) \times 10^2$	0.11
S3.1 (30 min 200°C)	(641.93 ± 0.07)	$(7.77 \pm 0.58) \times 10^2$	0.08
S4.1 C1	(25.08 ± 0.00)	$(3.94 \pm 0.16) \times 10^1$	0.04
S4.1 C5	(133.90 ± 0.02)	$(3.07 \pm 0.15) \times 10^2$	0.05
S2.1 after 400 (outlier)	$(6.60 \times 10^5 \pm 7.65 \times 10^2)$	$(6.60 \pm 0.75) \times 10^5$	0.11

Table 3: Resistivity and resistance values with propagated uncertainties for measured samples,

Thickness uncertainties from profilometry dominate the total error. Sample S2.1 after 22°C annealing procees (recorded in volts) remains an outlier.

4.3 Optical properties

In order to analyze the the transparency of our samples, an optical spectrometer was used as explained in section 3.2. The results of the Transmittance and Reflectance measurements are displayed in Fig. 5,6.



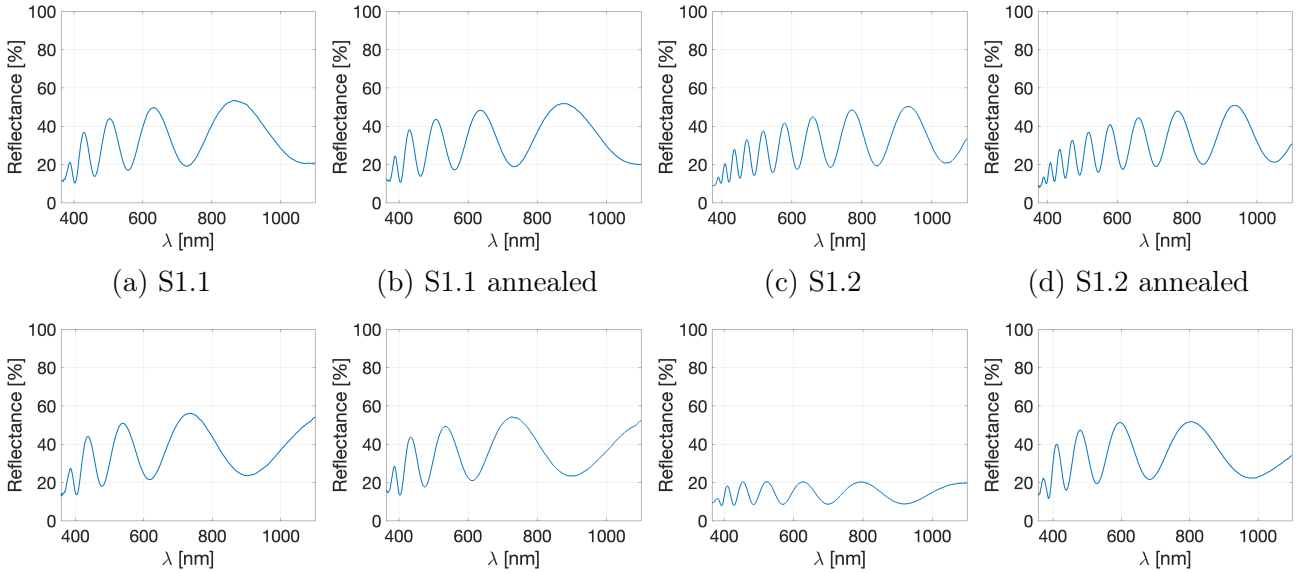
(a) S1.1 (b) S1.1 annealed (c) S1.2 (d) S1.2 annealed

(e) S2.1 (f) S2.1 annealed (g) S3.1 (h) S4.1

Figure 5: Transmittance plots for samples S1–S4.

From these results it is visible that our samples are mostly transparent for wavelengths above 400 nm, with a Transmittance that oscillates around 80%, and a Reflectance that oscillates around 30%.

From the Transmittance data and using Eqs. (11),(12) the absorption coefficient $\alpha(\lambda)$ as well as the Tauc plot were obtained. The overall trend of these results is similar for all samples

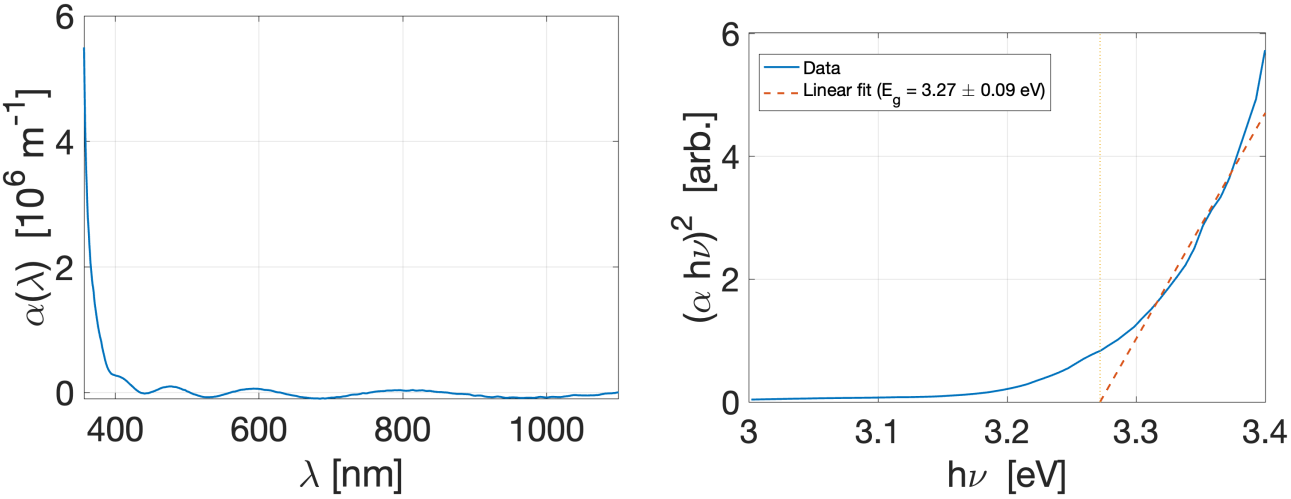


(a) S1.1 (b) S1.1 annealed (c) S1.2 (d) S1.2 annealed

(e) S2.1 (f) S2.1 annealed (g) S3.1 (h) S4.1

Figure 6: Reflectance plots for samples S1–S4.

and it is shown in Fig.7, using sample S4.1 as an example. The rest of the estimated band gaps as well as the estimated film thickness using Eq. (13) are presented in Tab. 4.



(a) (b)

Figure 7: (a) Absorption coefficient and (b) Tauc plot of sample S4.1.

Sample	S1.1	S1.1 Annealed	S1.2	S1.2 Annealed
Optical gap ΔE_g [eV]	3.27 ± 0.08	3.3 ± 0.1	3.2 ± 0.1	3.3 ± 0.2
Thickness d [nm]	620 ± 10	610 ± 10	1140 ± 20	1160 ± 20

Sample	S2.1	S2.1 Annealed	S3.1	S4.1
Optical gap ΔE_g [eV]	3.3 ± 0.1	3.3 ± 0.2	3.3 ± 0.2	3.27 ± 0.09
Thickness d [nm]	520 ± 10	530 ± 10	750 ± 10	580 ± 10

Table 4: Optical gap values ΔE_g obtained through the Tauc plot.

It is visible that the band gap values are very similar between all samples : around 3.3 Ev. Thickness results range from 500nm to 750nm for the half hour deposition samples and go up to 1.1μm for the 1h deposition, almost double the thickness.

These results will be discussed in further detail in section 5.

4.4 X-ray diffractometry

Results of the X-ray measurements are shown in Fig. 17. Knowing that the wavelength of the incident X-ray beam is $\lambda = 1.54056$ Å, the crystallite size D can be calculated after determining θ and β as shown in the example given in Fig. 8. The results are reported on Tab.5. The X-ray measurement was not performed for sample S2.1 before annealing. However, similar results to those of sample S1.1 are expected, since both were deposited under the same conditions.

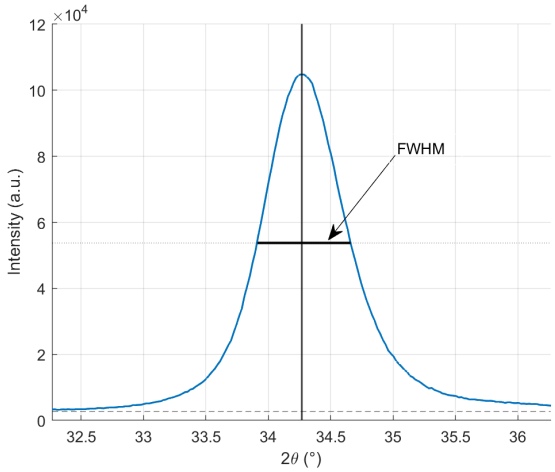


Figure 8: Determination of the FWHM on circle C5 of sample S4.1

Sample	S1.1	S1.1 Annealed	S1.2	S1.2 Annealed
Crystallite size D [nm]	7.2 ± 0.8	7.7 ± 0.9	8.4 ± 0.4	6.7 ± 1.4

Sample	S2.1 Annealed	S3.1	S4.1.C1	S4.1.C5
Crystallite size D [nm]	14 ± 4	7.9 ± 0.6	16.2 ± 0.4	11.2 ± 0.3

Table 5: Crystallite size of different samples obtained through X-ray diffractometry.

5 Discussion

Profile measurements

Firstly, it can be observed from the results in Tab. 2 that the deposition time increases the sample thickness, since sample S1.2, which was deposited under the same conditions as sample S1.1 but for one hour instead of 30 minutes, exhibits a thickness almost twice as large.

Secondly, both the deposition and annealing temperatures appear to have a strong positive effect on the surface roughness quality. Indeed, the roughness values of each deposition after annealing are lower than before, while the samples deposited at elevated temperature also show significantly improved roughness compared to those deposited at room temperature. This can also be qualitatively observed in Fig. 13, where, for instance, the profile of sample S2.1 after annealing appears much flatter than the profile of S2.1 before annealing.

Unfortunately, no clear conclusion can be drawn regarding whether heating the sample during or after deposition yields better results. Although the roughness value of sample S4.1 in circle C1 is the lowest among all, the value of the same sample in circle C5 is higher than that of sample S2.1 after annealing. To reach a more reliable conclusion, it would be necessary to analyze multiple samples in order to reduce the margin of error.

5.1 Resistivity measurement

Effect of annealing. Annealing in air had a strong positive effect on conductivity . For S1.1, resistivity decreased from 4.9×10^3 to 2.8×10^2 $\Omega\,\mathrm{m}$ after annealing, while for S1.2 it decreased from 1.2×10^2 to 9.1×10^1 $\Omega\,\mathrm{m}$. At the same time, surface roughness was reduced (for example, R_q dropped from 0.147 to 0.128 µm for S1.1, and from 0.246 to 0.121 µm for S1.2). These changes indicate that annealing promotes recrystallization and grain coalescence, improving crystal quality and reducing electron scattering at grain boundaries. The enhanced structural order is therefore directly responsible for the observed increase in conductivity.

Effect of substrate heating during deposition. Samples S2.1, S3.1, and S4.1 were deposited with the substrate heated to 200 °C or more. These films are thinner and smoother than the S1 series deposited at room temperature. Their roughness values are below 0.05 µm, and their resistivities range from 10^2 to 10^3 $\Omega\,\mathrm{m}$, which confirms that substrate heating leads to more uniform and better-crystallized ZnO layers. The elevated temperature increases the mobility of adatoms on the surface, allowing them to arrange into a denser and more ordered structure. The best result is obtained for sample S4.1 C1, which combines the lowest roughness (0.016 µm) with the lowest resistivity (3.9×10^1 $\Omega\,\mathrm{m}$). This strong correlation demonstrates that smoother and denser films exhibit superior conductance.

5.2 Optical properties

Firstly, it is visible in Fig. 5 that Transmittance varies little across all samples. A notable exception is sample S1.2 (Fig. 5c, 5d), whose spectrum shows more pronounced fringes (peaks and valleys). This is consistent with its longer deposition time (1 h), which yields a thicker film (confirmed in Tab. 4). A thicker layer satisfies the interference boundary conditions for a larger set of wavelengths, hence the higher density of oscillations.

It is also instructive to compare the optical response before and after annealing. Figure 14 summarizes these Transmittance comparisons. For short deposition times, post-deposition thermal treatment has little visible impact (Fig. 14a, 14c). In contrast, for the longer deposition time the effect is more apparent (Fig. 14b) and thermal treatment seems to improve transmittance.

Another relevant comparison is between S1.1 and S2.1 after annealing, to assess the effect of a higher annealing temperature. An analogous comparison between S3.1 and S4.1 probes the effect of different *in situ* heating temperatures. These results are shown in Fig. 15. At first glance, annealing at 400°C appears to yield better Transmittance than at 200°C (Fig. 15a). However, this apparent difference cannot be attributed to annealing alone: both S1.1 and S2.1 show almost no improvement from before to after annealing (Fig. 14b, 14c). The difference seen in Fig. 15a therefore most likely reflects superior deposition quality in sample S2.1 rather than a genuine temperature-induced enhancement.

For S3.1 and S4.1, Fig. 15b does suggest that a higher *in situ* heating temperature improves Transmittance. This conclusion holds only if all other deposition parameters are comparable. Given that early runs were performed with less experience, it is plausible that the improvements in process control that were made up to samples S3.1 and S4.1 made the samples comparable in this context. To reach firmer conclusions, additional measurements are needed—for example, dividing a single sample and applying different annealing treatments to each half. For *in situ* heating, a larger set of samples prepared under otherwise identical conditions should be characterized to ensure that the superior performance of S4.1 over S3.1 is not coincidental.

Additionally, an important limitation is visible in Fig. 5 and Fig. 6: for most samples the sum of Transmittance and Reflectance exceeds 100% (see Fig. 16 in the annex), which is unphysical and indicates a measurement error. This occurs for all samples except S3.1 (Fig. 16a). Because the Transmittance traces are comparatively stable—oscillating around a roughly constant level—the most likely culprit is the Reflectance data. Reflectance was remeasured repeatedly, attempting to reproduce the S3.1 behavior, but without success. The root cause remains unclear. If the Halogen lamp were at fault, Transmittance would likely be affected as well. Conversely, if the issue stemmed from handling during the measurements, it is surprising that repeated trials did not eliminate it. Further controlled tests will be required to isolate the source of this discrepancy.

Finally, to comment on the results in Tab. 4, the obtained band gap values are well within what the literature indicates which is $\Delta E_g = 3.276 \pm 0.033$eV [6]. The obtained values have then a relative error of at most 7% (assuming maximum error for sample S2.1 annealed). As for the thickness results, by comparing the data obtained with the profilometer and the one obtained with the optical spectrometer we obtain Tab. 7 (see annex). There it is visible that the difference in measured thickness rarely descends from 20%, indicating that either Eq. (13) is not completely fit for this scenario (might not take into account the glass substrate) or simply the transmittance data is not precise enough.

5.3 X-ray diffractometry

The crystallite size values presented in Table 5 show how both deposition and annealing conditions influence the crystalline structure of the films. Firstly, a longer deposition time appears to slightly increase the crystallite size, as sample S1.1 shows a value of 7.2 ± 0.8 nm, while sample S1.2 reaches 8.4 ± 0.4 nm before annealing. This trend is reasonable, since a longer deposition time allows more atoms to accumulate on the substrate surface, allowing the clusters to grow bigger and the atoms to move more easily, which makes the crystallite size slightly larger.

In addition, the crystallite size generally increases either after annealing or when the deposition is carried out at a higher temperature, which was the expected behavior [7]. A temperature of 200 °C appears to improve the crystal quality, although the effect is generally modest. For instance, sample S1.1 shows a slight increase from 7.2 ± 0.8 nm to 7.7 ± 0.9 nm, while sample S3, deposited in situ at 200 °C , reaches 7.9 ± 0.6 nm. However, in the case of sample S1.2, the crystallite size even decreases from 8.4 ± 0.4 nm to 6.7 ± 1.4 nm, although this value should be interpreted with caution due to the relatively high uncertainty. However, for annealing or deposition at 400 °C, the results show a significant improvement, yielding crystallite sizes of 14 ± 4 nm for sample S3.1 and 16.2 ± 0.4 nm and 11.2 ± 0.3 nm for circles C1 and C5 of sample S4.1, respectively.

These results suggest that there is an optimal temperature for crystal formation, which appears to be much closer to 400 °C than to 200 °C.

Finally, the results are likely to be somewhat inaccurate, since the influence of the glass substrate was completely neglected in the calculations. As shown in the X-ray diffraction spectrum of the glass in Fig. 17, there are certain angles, particularly at lower values, where the contribution of the glass to the measured signal does not appear to be negligible.

5.4 Deposition quality and possible protocol improvements

The deposition quality across samples was found to be highly variable. Significant heterogeneity in the deposited layer was observed even within single samples. Only subsamples *C.1* and *Sq.1* exhibited a sufficiently uniform surface morphology, within each sample, to be considered homogeneous for subsequent analysis. It can be noted samples S0.1 , S0.2 , S5.1 and S5.2 were not used for the anlaysis since the deposition quality was esteemed poor.

A major factor influencing this variability is attributed to imperfections in the cache(sample holder). The surface was not entirely planar, resulting in non-uniform electric field distribution and, consequently, uneven sputtering rates across the substrate. For future experiments, it would be advisable to design a new cache system ensuring both flatness and mechanical stability of the glass substrates during deposition. This could be achieved by using a machined metallic support or an adjustable holder cache bending sample tilt.

Moreover, maintaining a high level of cleanliness proved to be essential. Samples cleaned with dry, pressurized air before insertion into the chamber exhibited visibly improved uniformity and adhesion. Contaminants such as dust or residual moisture can significantly affect the nucleation and growth of ZnO films, thus altering their optical and electrical properties.

Finally, only the best-quality samples—those showing visually uniform deposition—were selected for further characterization. Detailed observations on the substrate conditions, chamber pressure evolution, and specific anomalies during deposition are reported in the experimental logbook (see Lab Notes [8] for full details).

6 Conclusion

The objective of the study is to set out to relate deposition parameters—time, *in situ* substrate heating, and post-deposition annealing—to the structural, electrical, and optical properties of sputtered ZnO films on glass. The results consistently show that increasing the thermal exposure improves film quality. Doubling the sputtering time from 30 minutes to 1 hour nearly doubled the thickness, confirming a steady deposition rate, and the thicker film exhibited the expected denser interference fringes in transmission. Heating, whether applied during growth or via annealing, reduced surface roughness and enhanced electrical performance: in the S1 series, annealing drove the resistivity down by more than an order of magnitude, and the best simultaneous roughness–conductivity trade-off was obtained for films grown at elevated substrate temperature (notably S4.1 C1) where resistivity was even 1 order of magnitude lower than all other samples.

Optically, all samples remained largely transparent above roughly 400 nm. Tauc analysis yielded a direct band gap close to 3.3 eV for every processing route, in line with canonical ZnO values and indicating that neither longer deposition nor additional thermal treatment compromised the fundamental electronic structure. Thickness estimates derived from interference fringes followed the correct trends but deviated from profilometry by about 20–30 % in many cases, suggesting either model simplifications for our substrate/film stack or residual calibration errors in the spectra. X-ray diffraction indicated nanocrystalline domains in the 7–16 nm range and generally larger crystallites at higher temperatures, consistent with improved crystallinity under stronger thermal budgets, though possible glass-substrate contributions at low angles advise caution in the most affected regions.

A persistent caveat in the optical dataset is that $R + T$ exceeded 100 % for most samples, pointing to a systematic issue in reflectance measurement or referencing rather than in transmission, which appeared comparatively stable. While this limitation does not overturn the observed process–property trends, it constrains absolute optical quantification and should be addressed by revisiting the reflectance baseline, reference standard, and geometry.

In answer to the objective stated in the Introduction, we find that ZnO films intended for optoelectronic applications benefit from higher thermal budgets—either through *in situ* heating up to 400°C or through post-deposition annealing—combined with deposition times tailored to the target thickness. These conditions yield smoother, better-crystallized layers with substantially lower resistivity, while preserving a wide, direct band gap and high visible transparency. Future work should prioritize a flatter and more stable sample holder, stricter cleanliness, a re-established reflectance calibration, and split-sample protocols with complementary metrology (e.g., Hall measurements and spectroscopic ellipsometry) to transform the strong trends reported here into a robust process window for optimized ZnO thin films.

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7 Annexes

Used parameters and uncertainty on measured quantities

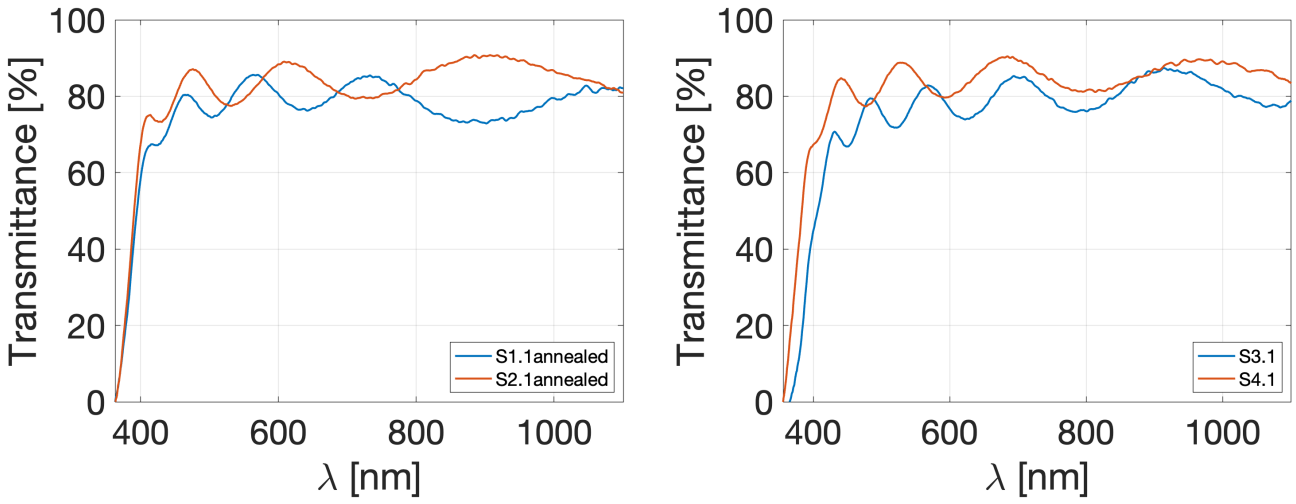
Table 6: Measured quantities and associated uncertainties.

Quantity	Symbol	Unit	Δ
Voltage	V	mV	0.02
Current intensity	I	µA	0.02
Wavelength (optical spectrometer thickness measurement)	λ_i	nm	5



Figure 9: Experimental setup for optical measurements

Figures



(a) (b)

Figure 15: Comparison of Transmittance values between heat treatment at 200ºC and 400ºC for (a) annealed and (b) *in situ* heated samples



Figure 10: Whole magnetron machine with vacuum chamber

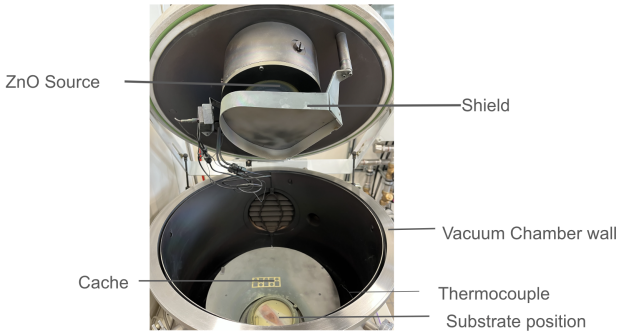


Figure 11: View of the vacuum chamber from closer, seen in Image 10

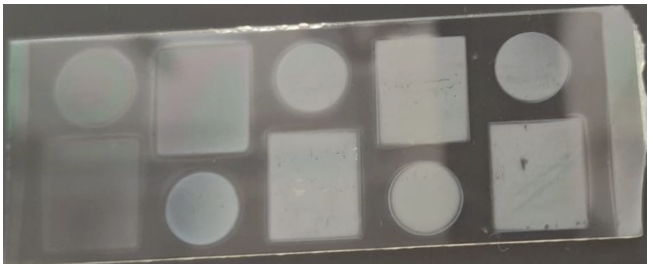
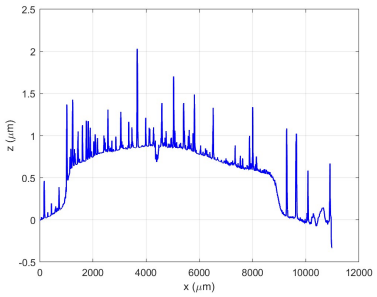
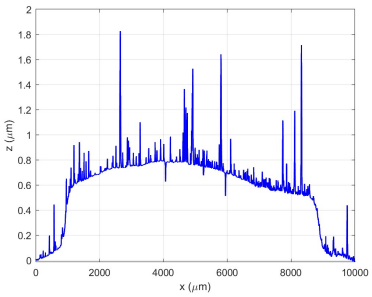


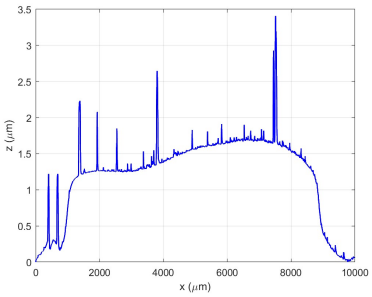
Figure 12: Sample S4.1, 30-minute deposition at 400 °C.



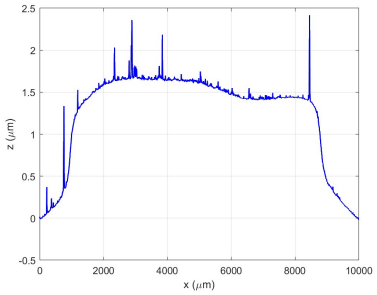
(a) S1.1



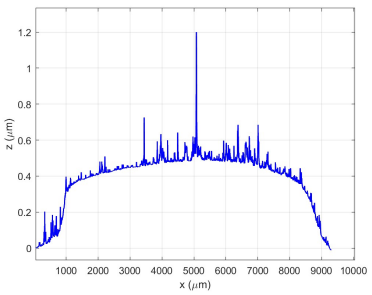
(b) S1.1 annealed



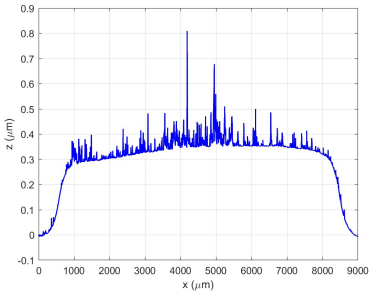
(c) S1.2



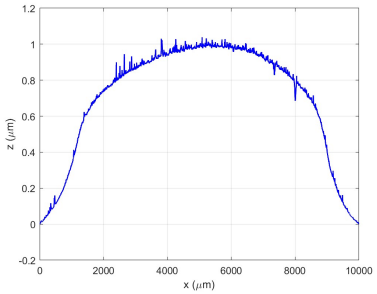
(d) S1.2 annealed



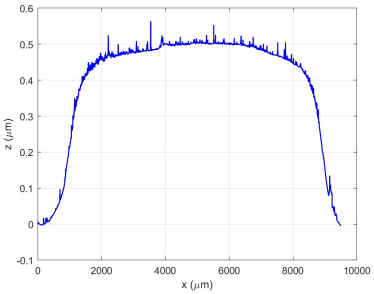
(e) S2.1



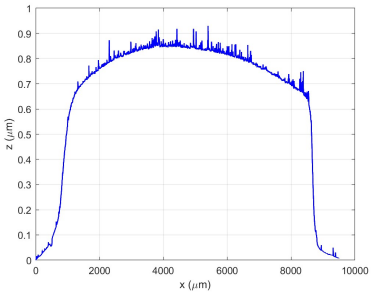
(f) S2.1 annealed



(g) S3.1

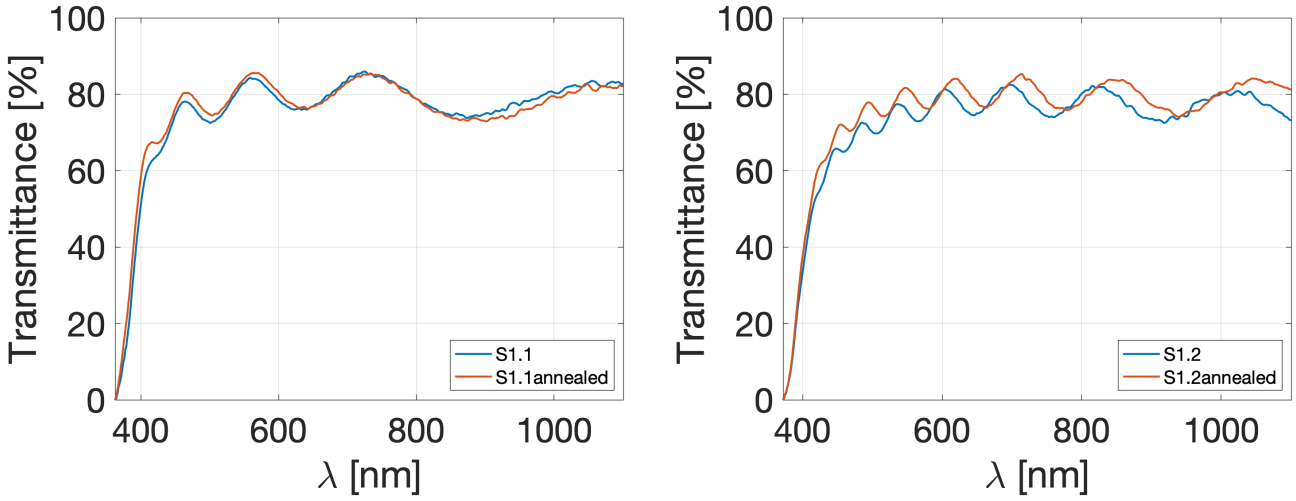


(h) S4.1.C1

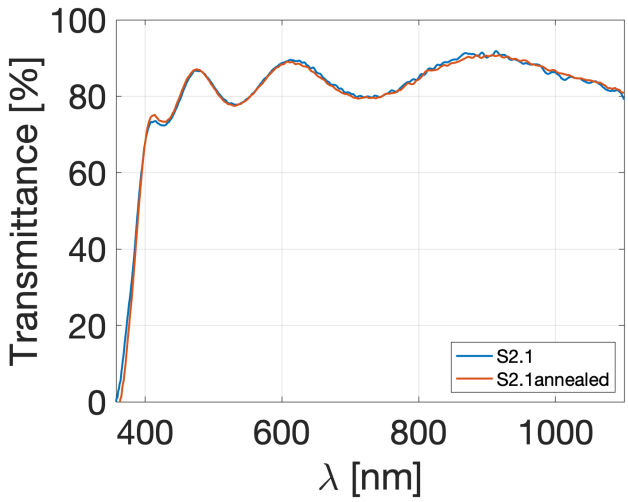


(i) S4.1.C5

Figure 13: Surface height profiles of samples S1–S4 under different deposition and annealing conditions.

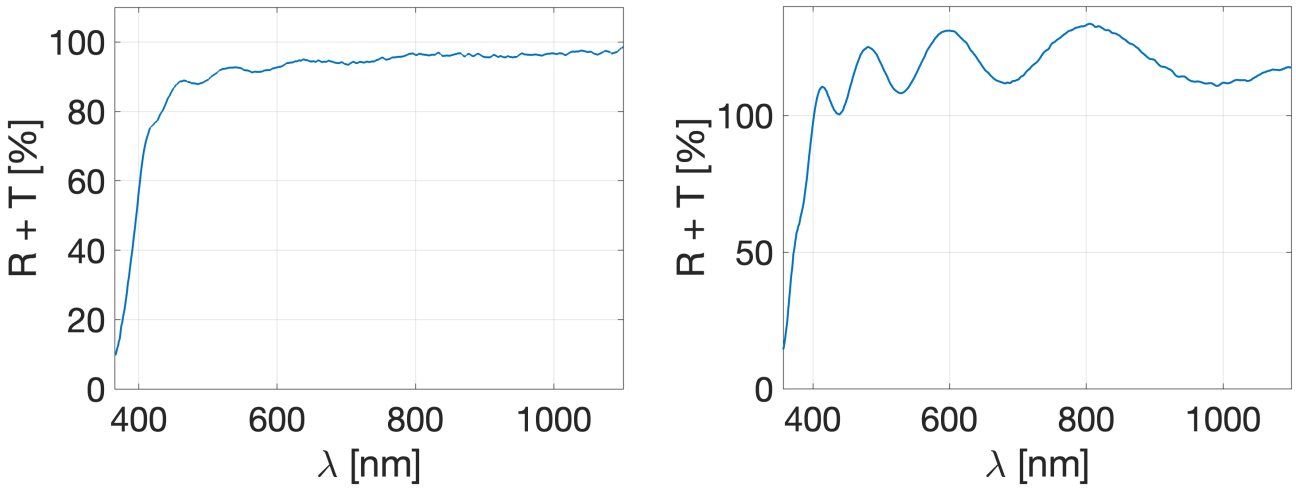


(a) (b)



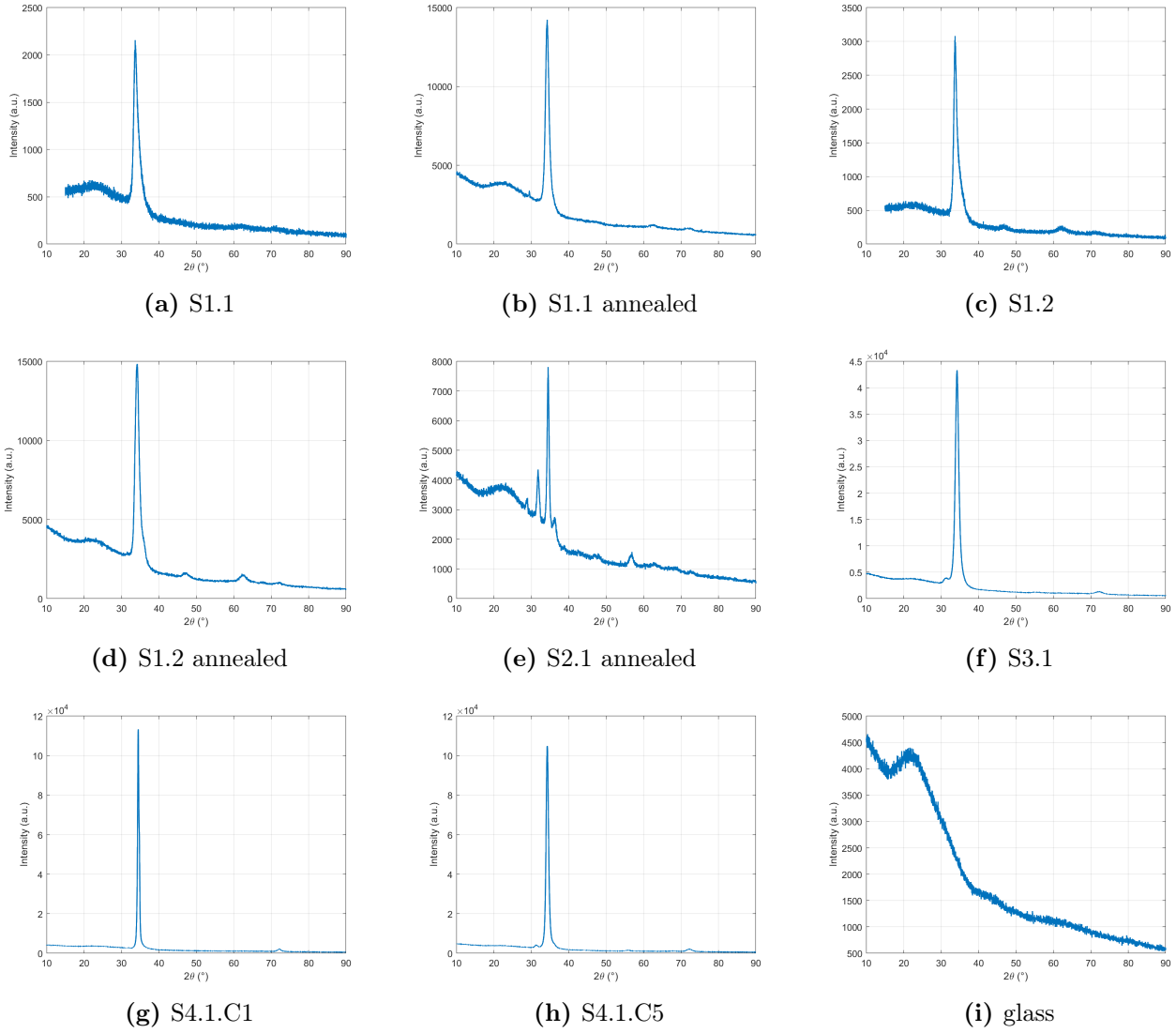
(c)

Figure 14: Comparison of Transmittance values for samples (a) S1.1 , (b) S1.2 and (c) S2.1 before and after annealing.



(a) (b) Sample S4

Figure 16: Sum of reflectance and transmittance $(R + T)$ for samples (a) S3.1 and (b) S4.1



(a) S1.1 **(b)** S1.1 annealed **(c)** S1.2

(d) S1.2 annealed **(e)** S2.1 annealed **(f)** S3.1

(g) S4.1.C1 **(h)** S4.1.C5 **(i)** glass

Figure 17: X-ray diffraction spectra of samples S1–S4 under different deposition and annealing conditions, along with that of the glass substrate.

Table 7: Thickness comparison: profilometer vs optical spectrometer.

Sample	Profilometer d [µm]	Optical d [µm]	Δ (Opt. $-$ Prof.) [µm]	$\lvert\Delta\rvert/d_{\text{prof}}$ [%]
S1.1	0.83 ± 0.15	0.620 ± 0.010	-0.210	25.3
S1.1 Annealed	0.74 ± 0.13	0.610 ± 0.010	-0.130	17.6
S1.2	1.57 ± 0.24	1.140 ± 0.020	-0.430	27.4
S1.2 Annealed	1.61 ± 0.11	1.160 ± 0.020	-0.450	28.0
S2.1	0.48 ± 0.05	0.520 ± 0.010	0.040	8.3
S2.1 Annealed	0.35 ± 0.04	0.530 ± 0.010	0.180	51.4
S3.1	0.94 ± 0.07	0.750 ± 0.010	-0.190	20.2
S4.1.C1	0.49 ± 0.02	0.580 ± 0.010	0.090	18.4
S4.1.C5	0.82 ± 0.04	0.580 ± 0.010	-0.240	29.3

Error propagation

Uncertainty propagation formula: The following formula was used to determine some of the uncertainties of the calculated values:

$$\Delta G = \sum_{i=1}^{n} \left| \frac{\partial G}{\partial x_i} \right| \Delta x_i \tag{14}$$

where n is the number of variables of G.

Uncertainty on thickness (optical spectrometer) :

$$\Delta d = \frac{1}{2n_1} \frac{\lambda_2^2 \Delta\lambda_1 + \lambda_1^2 \Delta\lambda_2}{(\lambda_2 - \lambda_1)^2}. \tag{15}$$

Uncertainty of thickness (profilometer) : This error is estimated as the standard deviation of the sample's thickness.

$$\Delta t = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (z_i - \bar{z})^2} \tag{16}$$

Uncertainty of roughness : It was estimated using a block bootstrap method, which accounts for the spatial correlation of the profile and provides a more realistic evaluation of the error than analytical approaches.

Uncertainty of conductivity From Ohms Law

$$R = \frac{V}{I} \quad \Rightarrow \quad \frac{\Delta R}{R} = \sqrt{\left(\frac{\Delta V}{V}\right)^2 + \left(\frac{\Delta I}{I}\right)^2}. \tag{17}$$

For conductivity the error propagation becomes :

$$\frac{\Delta \rho}{\rho} = \sqrt{\left(\frac{\Delta V}{V}\right)^2 + \left(\frac{\Delta I}{I}\right)^2 + \left(\frac{\Delta t}{t}\right)^2}. \tag{18}$$

The whole used uncertainties are :

Uncertainty on crystallite size :

$$\Delta D = \left| \frac{K\lambda}{\beta^2 \cos\theta} \right| \Delta\beta + \left| \frac{K\lambda \sin\theta}{\beta \cos^2\theta} \right| \Delta\theta \tag{19}$$

where $\Delta\beta$ and $\Delta\theta$ are estimated numerically.

Sample	R [Ω]	ρ [Ω m]	$\Delta R/R$	$\Delta t/t$	$\Delta\rho/\rho$	$\Delta\rho$ [Ω m]
S1.1 (30 min)	2743.23	4.89×10^3	1.148	0.181	1.162	5.68×10^3
S1.1 Annealed	175.53	2.77×10^2	9.165	0.176	9.167	2.54×10^3
S1.2 (1 h)	116.60	1.17×10^2	17.155	0.153	17.156	2.00×10^3
S1.2 Annealed	91.52	9.15×10^1	22.903	0.068	22.904	2.10×10^3
S2.1 (Before ann.)	6218.66	1.56×10^3	0.611	0.104	0.620	9.68×10^2
S2.1 Annealed	277.08	2.77×10^2	4.220	0.114	4.222	1.17×10^3
S3.1 (30 min 200°C)	641.93	7.77×10^2	2.850	0.075	2.851	2.22×10^3
S4.1 C1 (Square 1)	25.08	3.94×10^1	17.330	0.041	17.331	6.82×10^2
S4.1 C5 (Square 4)	133.90	3.07×10^2	2.680	0.049	2.681	8.23×10^2
S2.1 after 400	660401.00	6.60×10^5	1.003	0.114	1.009	6.66×10^5

Table 8: Resistivity measurement propagation values and propagated uncertainties for all measured samples. Electrical uncertainties use $\Delta V = \pm 3$ mV and $\Delta I = \pm 2$ μA;